MobilePixorNeXt

hist14 BEV + Focal Context · 666,768 backbone parameters (including neck)

Backbone

Bottom-up encoder

hist14 input

14 × 800 × 704

Stem: Conv 3×3, 14→32, s=2
Conv 3×3, 32→32, s=1 · BN + SiLU

32 × 400 × 352 /2

Conv 3×3, 32→48, s=2 — BN + SiLU

Stage 2 / C3 · 3 NeXt blocks

NeXt | NeXt | NeXt

48 × 200 × 176 /4

Conv 3×3, 48→96, s=2 — BN + SiLU

Stage 3 / C4 · 4 NeXt blocks

NeXt | NeXt | NeXt | NeXt | Focal

96 × 100 × 88 /8

Conv 3×3, 96→128, s=2 — BN + SiLU

Stage 4 / C5 · 2 NeXt blocks

NeXt | NeXt

128 × 50 × 44 /16

L5: Conv 1×1
128→48

48 × 50 × 44

Top-down SG-FPN

Conv 3×3, 32→32
BN + SiLU

32 × 200 × 176

32 channels
stride 4

G **P3**

Gated fusion

L3: Conv 1×1
48→32

32 × 200 × 176

Bilinear ↑2
Conv 3×3, 48→32
BN + SiLU

G **P4**

Gated fusion

L4: Conv 1×1
96→48

48 × 100 × 88

Bilinear ↑2
DWConv 3×3, 48→48
BN + SiLU

Detection heads

Per branch: 3×3 + 3×3 + 1×1

Conv	Heatmap 200×176×K
Conv	Offset 200×176×2
Conv	Size 200×176×2
Conv	Yaw 200×176×2
Conv	IoU quality 200×176×1

Independent template per group
Car: K=1; Pedestrian+Cyclist: K=2

Feature refinement

C3 / stride 4: convolutional features

3 NeXt blocks → C3

C4 / stride 8: Focal Context

BN → Conv1×1 → Q, S, G (D=64)

DW3×3 d=1 → DW3×3 d=2 → DW3×3 d=3

Sequential local levels + global mean

Softmax(G) → context mix → Conv1×1

Q ⊙ context → projection + BN

Y = X + γ · residual, $\gamma_0 = 0.001$

(A) MobilePixorNeXt block

Depthwise spatial mixing → pointwise channel expansion → residual

X → DW 7×7 C→C → BN → PW 1×1 C→2.5C → SiLU → PW 1×1 2.5C→C → BN → γ → + → Y

Identity shortcut

Y = X + γ ⊙ F(X) · γ is learned per channel; initialized to 10^{-5}

(B) Scale-gated fusion

L: lateral feature U: upsampled feature

L → + → DWConv 3×3 zero-init → 2 · sigmoid → × → + → P

U → +

P = U + 2σ(DWConv(L + U)) ⊙ L

At initialization: gate = 1, so P = U + L

Shapes use C × H × W (batch omitted). G = scale-gated fusion. Cuboid dimensions are schematic.